

00:00

Yeah. Good morning, everyone. Thank you for joining me in week nine. Today's topic is about making prediction. You know, in all previous weeks, you were talking about different, you know, aspects of machine learning, data science. You learned how we can clean data, deal with missing data, duplicating data, how to do visualization, and all these small pieces. Today, you do something which is actually the end of the pipeline. So you will see how different pieces fall into right place. It makes a better sense, and you will have a big picture of what we mean by a machine learning model. Okay? So today's coding is not complicated, but it's mainly about some concepts that I'm going to explain. Then I go to the code, and you will find it pretty interesting, I believe. Week 10 and 11, we don't have any coding. They are mainly about some concepts. It's mainly about human-centered AI, NLP, and some other AI contemporary concepts. So today is the last coding that we have in this unit. So let's talk about making prediction.

01:31

Today's agenda is about talking about correlation. Then you will see how you can visualize them using a scatter plot and heat map. Then we talk about some models that we use for prediction. The simplest one is regression. We have different regression like linear, multiple linear, polynomial, logistic regression, and least regression. They are very similar. There are some subtle differences among them. I'll explain in details. Then you get familiar with a new library in Python, Scikit-learn, or sometimes we call it SciLearn. Then, again, Seaborn is being used for visualization, as you see. We have two new functions, LM plot, which stands for linear model plot, and residue plot, which is for residuals. So let's see what we have.

02:37

The first topic is correlation between variables. Technically, it's between two variables, but you can expand it for many different variables. As you see here, correlation coefficient is simply a number or a metric that is being used to measure strengths and direction between two variables. These variables are supposed to be numerical, both. So two columns that have numeric values can be used to calculate correlation coefficient. And by numeric, I mean any number, decimal or any number.

03:16

you you I hope you can hear me now.

08:04

Yep, you're back. Yeah, sorry for Tim's problem. Okay, where did you lose me? I hope you can hear me now. Yep, you're back. Yeah, sorry for Tim's problem. Okay, where did you lose me? Here? Did you hear this slide, guys? A statistical measure? Okay, let me go through that quickly. Here. I told you that another name that we have for correlation coefficient is r-value. And as I mentioned, it only considers a linear relationship between two variables. It doesn't consider any nonlinear correlation, okay? For those relations, we have different measures or metrics.

08:48

So let's continue what we really mean by this r-value. Any number that is close to 1 indicates that there are a strong relationship, and same goes for negative one. So assume that we have negative one and positive one at the end of a spectrum. When relationship coefficient or correlation coefficient close to either of these two ends, we mean these are strongly correlated, okay? And when they get close to zero from both ends, the correlation coefficient shows less a strong relationship. OK, here, positive and negative simply mean that these relationships are reverse or positive. As you saw that when both increase, it's positive. When both decrease, it's also positive. They move in the same direction. But once they are moving in different direction, the relationship is negative. Let's say, as, for instance, age goes up, probably weights go up as well. So this is a positive correlation. But for negative correlation, I can think of another example. Let's say what we initially had. When the price goes up, the demand goes down. So they are moving in different direction, which shows negative correlation. So it doesn't have any other meaning, okay? negative correlation. So it doesn't have any other meaning, okay? So let's see. Here I have more, you know, clear example of correlation coefficient. It shows that if it's positive, they are moving in the same direction. And it can be any number from zero to positive 1 or negative 1. So since only shows the direction, if they are moving in the same direction or not. But the number, an absolute value of number, shows how strong is the relationship, okay?

10:58

So if these lines that you see, these dashed lines, guys, are showing what linear regression model gives you. Linear regression model simply gives you a line that predicts the relationship. And these dot points are your real data points. So the closer your dot points are to your line, it means the model is able to predict better. So when your dot points are scattered, OK, it doesn't show a good model. We want something like this or maybe like this. OK, so that's what you should bear in mind when it comes to correlation coefficient. And as I told you, another name for correlation coefficient is  $r$ -value.

11:54

But there is a delicate point here I want you to focus, guys. Look at these two curves at the top. These are two straight lines. And you know that curve is a general length for lines. So these two have the same  $R$  value or the same correlation coefficient, but you see the steep of these two lines are different. You know, one of them is close to horizontal line, but the other one is very steepy. I want you to bear in mind that a steepness or a slope is different from correlation coefficient. It doesn't mean that they are more correlated. It simply shows that, you know, this one. Let me see if I can get this laser. Yeah. This one, guys, this curve shows that the steep is more, which means when you move in  $X$  direction, you will have more change in one direction. Okay? Because the steep is higher. In here, as you see, when the value of  $X$  changes, let's say from two to four, the change in  $Y$  direction is much less. OK, so a steepness shows the rate of change. If it's more steep, it means the change that we have is stronger or more or faster. But it's totally different from correlation. So two lines with different steepness can have the same correlation. The same goes for these two ones. As you see, the correlation coefficient is low and same values. But steep is different, as you see. So bear in mind that a steep or a slope is totally different from correlation coefficient.

13:54

So another important point I want you to bear in mind is correlation is different from causation. Correlation is different from causation. So if I say by increasing temperature, the saline ice cream is increasing, I can see that they are correlated, right? They are moving in the same direction, so it shows positive correlation. But it doesn't mean that increasing temperature directly cause ice cream sales, okay? So please bear in mind that there are two different things. Correlation simply means two variables are moving together, but causation means one variable directly cause or impact the other on

e, okay? directly cause or impact the other one, okay? And that's something you may consider in some more complex situation. This is not the case of today's topic, but I wanted to put emphasis on that, okay?

15:06

So here is where we start talking about code. Let me close this and go to the code. You can come back later in here. So let's see what we have in your code for today. So guys, again, as you know, we first import the libraries that we need to work with. Okay.

15:28

Here I load a CSV data file, housing data. This data set talking about different features of a house, like size of living room, number of bedrooms, and also the price. of living room, number of bedrooms, and also the price. So I'm going to use this data set to predict the price of a house. That's the general purpose of this example. So I use read CSV because the extension of the file is CSV. And then I simply get the shape. As you know, it gives me the number of rows and number of columns. 18 columns. It's a bit too many, right? Maybe I don't need all of them. So I try to print what are these columns. Dot column is an attribute that gives you a list of the columns. These are 18 columns that I have. Date, price, bedroom, bathroom, and price, all information. So I do not need all of them, okay? So I'm going to simply pick a part of my data frame.

16:38

You've worked with this filtering procedure in previous weeks. So here I'm simply using query to filter some rows that are satisfying this condition. What are these conditions? I want to have the value of a square feet living column, which shows the size of living room, which is less than 8,000 square feet. And, okay, which means two conditions should be satisfied at the same time. Price should be less than a million. Okay. And another condition is price is positive, which means it's more than zero.

17:22

That's why here I was thinking, do we have any price which is negative or zero? So I tried to have a closer look at my data set. Here you see, I called my data set and checked the price column. That's column name. As you see here, column price is one of my columns, okay? So I look at this column, and I try to use this method, sort values.

When I use sort value, by default, it puts the value in column price in ascending order. So the smallest come first. And I pick the first five elements. the first five elements. As you see, the price value is zero, which means probably in my data sets, there were some missing values and I replaced those missing values by zero. So you see, all different things that you learned in previous weeks are coming into right place. So that's what happened after dealing with missing values. I don't want these zeros because they do not give me any information. That's why here I'm filtering them. I'm interested only on those rows of my data frame that the value of price is non-zero, something real, something meaningful. Okay, so from one dollar up to a million dollar.

18:48

OK, so from now on, this subset of my data frame is called housing and I'm using this as my reference. So what else I'm doing here? I'm creating a new column by giving a name, okay, has basement. I'm going to see if this part of my data set has basement or not. But how I do that? I simply look at this column of my data set, a square feet basement that shows the size of basement. Okay? So how I'm going to do that? I simply use this function, lambda. You remember lambda was an expression that we use to write one line or very simple functions. So what this function does, I should simply write the variable name that I have, then I put colon, then I write expression for my function. What I'm going to do, if the value of x is something more than zero, I say true, which means if the value of s-square-feet basement is positive or non-zero, it means it has a size. The basement has a size. So, yes, it has.